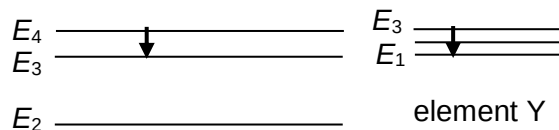


**28 C** Three spectral lines from Y means three energy levels for Y.

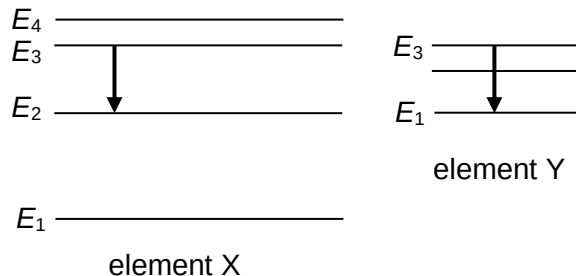
If  $\Delta E = E_4 - E_3$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will not be sufficient to excite X from ground to a higher level, resulting in 0 spectral line.



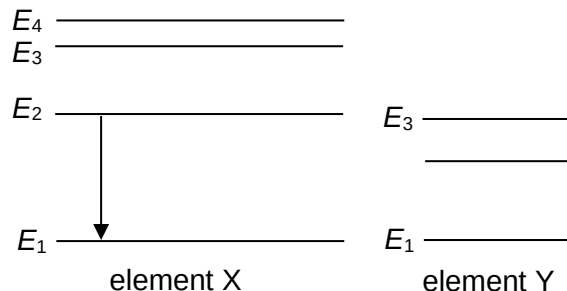
If  $\Delta E = E_4 - E_2$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will not be sufficient to excite X from ground to a higher level, resulting in 0 spectral line.



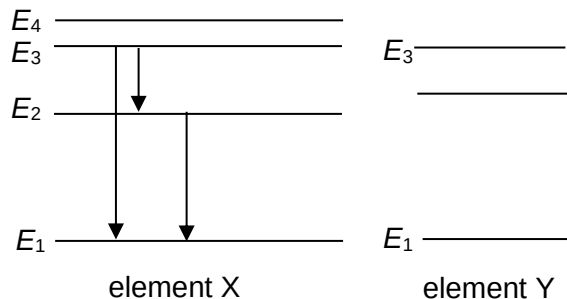
If  $\Delta E = E_3 - E_2$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will not be sufficient to excite X from ground to a higher level, resulting in 0 spectral line.



If  $\Delta E = E_2 - E_1$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will excite X to  $E_2$ , resulting in 1 spectral line.



If  $\Delta E = E_3 - E_1$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will excite X to  $E_3$ , resulting in 3 spectral lines.



If  $\Delta E = E_4 - E_1$  of X corresponds to  $E_3 - E_1$  of Y,  
then the same  $\Delta E$  will excite X to  $E_4$ , resulting in 6 spectral lines.

